

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

9747/02

Paper 2 Core Paper

16 September 2009

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
Total	

This document consists of **17** printed pages.



Section A
Answer **all** questions.

- 1 (a) Describe how DNA is organised into a chromosome.

.....

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.....

.....

.....

.....

[4]

Fig. 1.1 shows part of the HIV genome.

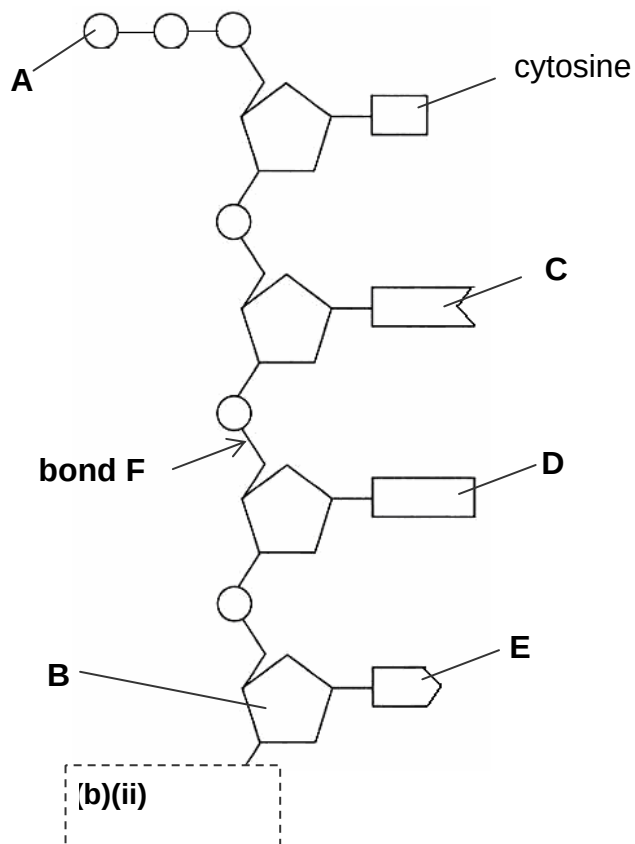


Fig. 1.1

(b)(i) Label structures **A** to **F**.

A - **D** -
B - **E** -
C - **F** - [3]

(ii) Write the functional group present in structure **B** in the box provided. [1]

(iii) Upon entering the host cell, the viral genome undergoes transcription before integration into the host genome.

On Fig. 1.1, illustrate the resultant product of this transcription from the viral genome shown. [2]

(iv) Name the enzyme used in this transcription.

..... [1]

Fig. 1.2 shows the structure of the nucleotide containing the base thymine and the structure of the drug AZT. This drug is used to prevent the replication of human immunodeficiency virus (HIV) inside human cells.

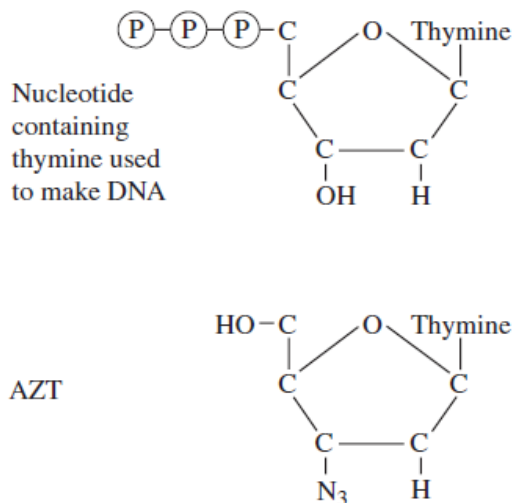


Fig. 1.2

(c) Suggest **one** way that AZT prevents replication of DNA from HIV.

..... [1]

[Total: 12]

2 Fig. 2 shows the life cycle of a bacteriophage.

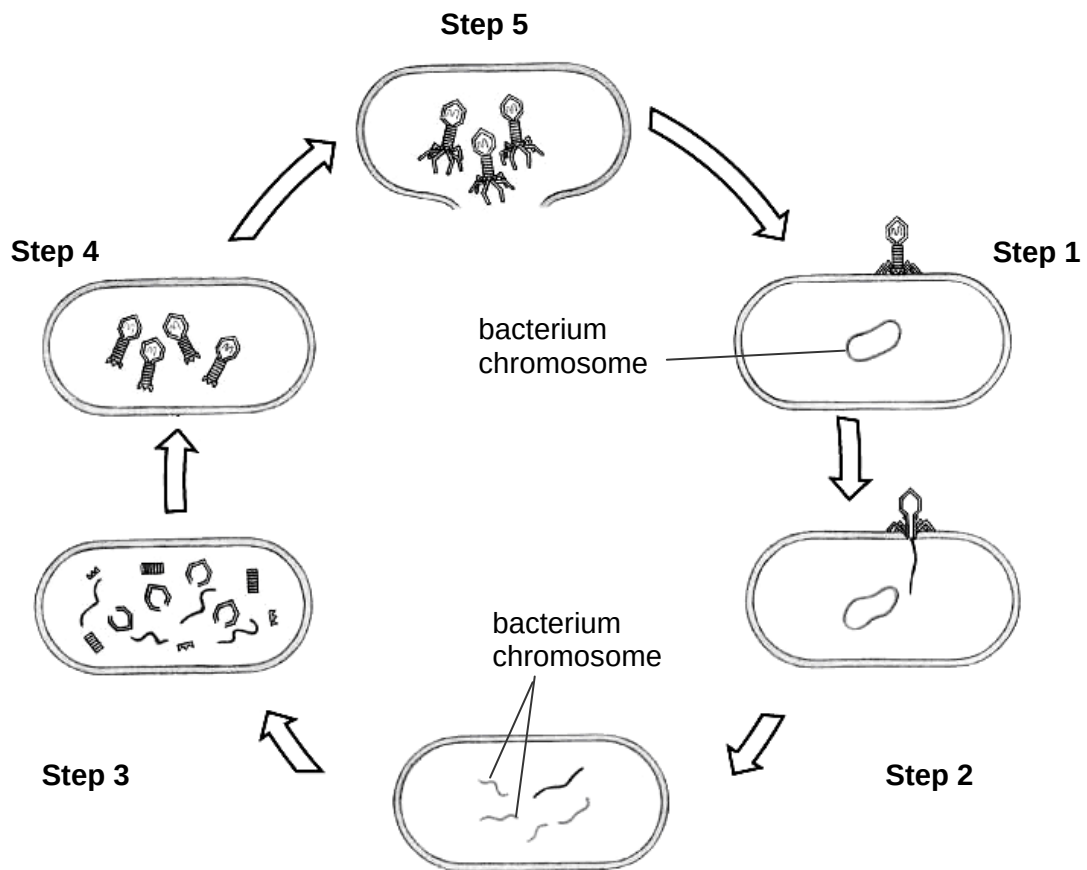


Fig. 2

(a) Describe events occurring in Steps 1 to 5.

Step 1

.....

Step 2

.....

Step 3

.....

Step 4

.....

Step 5

..... [5]

- (b) Suggest an identity of the phage shown.

..... [1]

- (c) Describe the role of phages in genetic transfer between bacteria.

..... [4]

[Total: 10]

3 Fig. 3.1 shows the *lac* operon in *E. coli*'s genome.

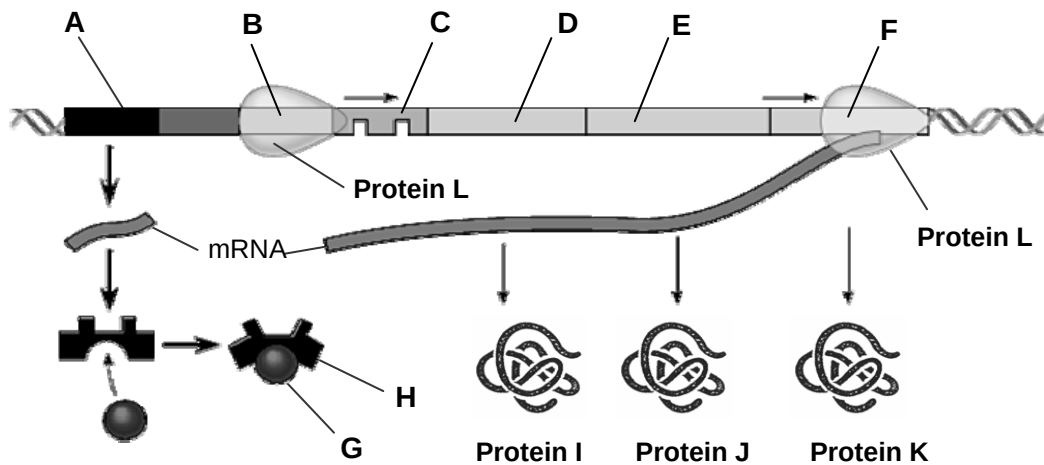


Fig. 3.1

(a)(i) Identify A – F of the DNA molecule.

A -	D -	
B -	E -	
C -	F -	[3]

(ii) Identify proteins I – L.

I -	K -	
J -	L -	[2]

(iii) Identify molecules G and H.

G -	H -	[1]
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(iv) Explain the effect of molecule G on the production of proteins I, J and K.

.....

.....

.....

.....

.....

[3]

Fig. 3.2 shows part of the base sequence of the mRNA resulting in the production of proteins I, J and K.

*

AUGAUAACU.....UAAAUGGAUAAA.....UGAAUG.....UAA
 5' 3'

Fig. 3.2

(b) With reference to Fig. 3.2,

(i) name the type of mRNA formed.

..... [1]

(ii) explain how three separate proteins may be synthesised from a single mRNA.

..... [1]

(iii) Describe the effect of a single base deletion at position indicated by * on lactose metabolism.

..... [2]

[Total: 13]

- 4 In the garden pea, *Pisum sativum*, the dominant alleles of two unlinked genes, **A/a** and **B/b**, are needed to make the pods tough and inedible due to lignin added to a thin cell layer. All other genotypes result in soft, edible 'sugar-snap' pea pods.

- Pods with genotypes including the dominant allele **A** have a thin layer of cells lining the pod.
- Pods with genotypes in which the recessive allele **a** is homozygous have **no** thin layer of cells lining the pod.
- Pods with genotypes including the dominant allele **B** have lignin added to the thin lining layer, when it is present, making the pods tough and inedible.
- Pods with genotypes in which the recessive allele **b** is homozygous do not have added lignin.

- (a) State the phenotypes of pea pods with the following genotypes:

AAbb

aaBB [1]

- (b) Two pea plants of genotypes **AABB** and **aabb** were interbred to give an F_1 generation and these in turn were interbred to give an F_2 generation.

Draw a genetic diagram of the cross, including gametes, showing the genotypes and phenotypes of the F_1 and F_2 generations. Give the ratio of phenotypes expected in the F_2 generation.

[5]

- (c)(i) Explain what is meant by the genes being 'unlinked'.

.....
..... [1]

- (ii) Suggest how the phenotypic ratio of the F_2 generation might change if the two genes are linked.

.....
.....
.....
..... [2]

[Total: 9]

- 5 In the kidney, glucose is freely filtered at the glomerulus and is reabsorbed in the proximal convoluted tubule. Two sodium-glucose co-transporters, SGLT1 and SGLT2, are responsible for glucose reabsorption.

SGLT1, which is also found in the gut and other tissues, accounts for about 10% of reabsorption. SGLT2, expressed exclusively in the proximal tubule, accounts for about 90% of reabsorption.

The concentration gradient that drives the action of these transporters is driven by the Na^+/K^+ ATPase pump and glucose is transported back into the blood via the GLUT2 glucose transporter.

Fig. 5.1 summarises the absorption of glucose.

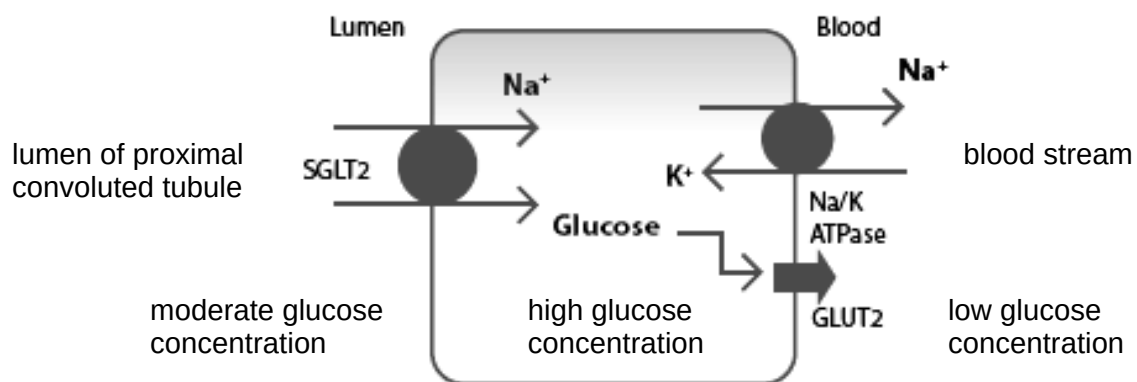


Fig. 5.1

- (a) With reference to Fig. 5.1, explain how glucose is reabsorbed from the lumen of the proximal convoluted tubule back into the blood stream.

.....

.....

.....

..... [2]

- (b) Explain how SGLT2 is expressed exclusively in the proximal tubule.

.....

.....

.....

..... [2]

Dapagliflozin, a selective SGLT2 inhibitor, is an experimental drug showing effective control in hyperglycaemia in type 2 diabetics. A placebo (PBO) was included where patients were given an inert compound which they were told was dapagliflozin. Fig. 5.2 below shows the results of a twelve week trial.

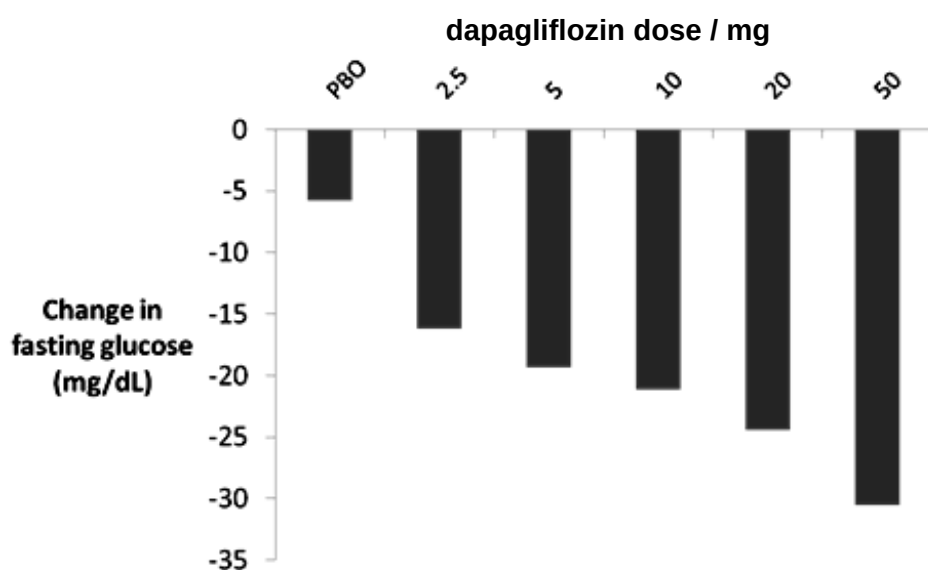


Fig. 5.2

(c)(i) Describe the effect of dapagliflozin on blood glucose level.

.....

.....

.....

..... [2]

(ii) Explain how dapagliflozin aids diabetics in lowering blood glucose level.

.....

.....

.....

..... [2]

(iii) Suggest what side effects dapagliflozin may have.

.....

..... [1]

(iv) Suggest why a placebo was included in the study.

.....
..... [1]

(d)(i) Suggest why SGLT2 instead of SGLT1 is targeted for drug design.

.....
..... [1]

(ii) Suggest why despite the large numbers of diabetic patients, diabetes was not a candidate for gene therapy.

.....
..... [1]

[Total: 12]

- 6 Fig. 6.1 shows part of a post synaptic membrane of a cholinergic synapse.

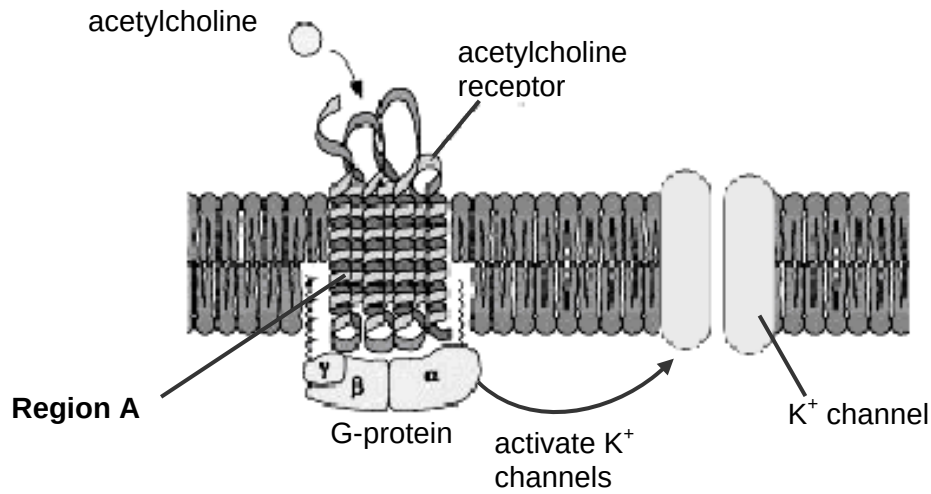


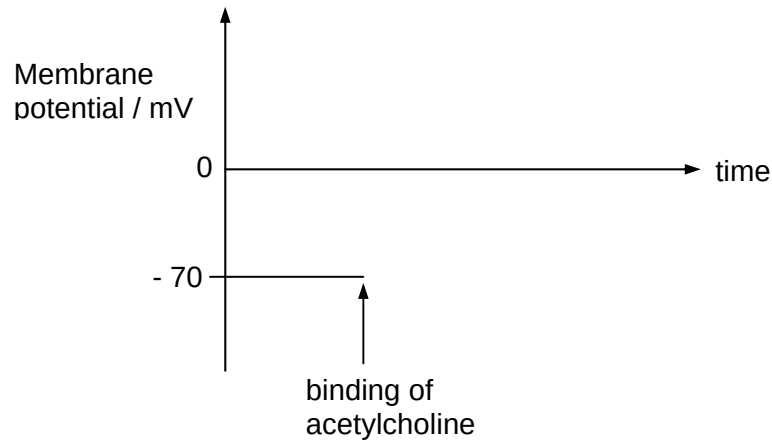
Fig. 6.1

- (a)(i) With reference to the post synaptic membrane potential, describe the events following the binding of acetylcholine to its receptor at the post synaptic membrane.

[3]

- (ii) On Fig. 6.1, illustrate with an arrow, the direction of K⁺ movement. [1]

- (iii) Sketch the changes in post synaptic potential in the axes provided below. [1]



- (b)(i) A mutation occurs in the α subunit of G-protein. The subunit loses its ability to hydrolyse GTP bound to it. Suggest a consequence of this mutation.

.....

.....

.....

..... [2]

- (ii) Explain what kind of mutation might lead to situation described in (b)(i).

.....

.....

.....

..... [2]

- (c) Describe the structure of **Region A**.

.....

.....

.....

..... [2]

[Total: 11]

- 7 Table 7.1 shows the amino acid differences in the alpha haemoglobin chains between various vertebrates.

	Human	Elephant	Platypus	Ostrich	Starling	Crocodile	Lungfish	Coelacanth	Goldfish	Shark
Human		26	40	43	41	47	83	70	68	71
Elephant			45	45	48	50	84	72	63	74
Platypus				54	52	51	89	74	70	76
Ostrich					26	36	91	75	68	73
Starling						47	91	77	67	70
Crocodile							85	78	70	77
Lungfish								90	94	86
Coelacanth									83	78
Goldfish										88
Shark										

Table 7.1

Fig. 7.2 shows a family tree based on differences between the alpha haemoglobins in Table 7.1

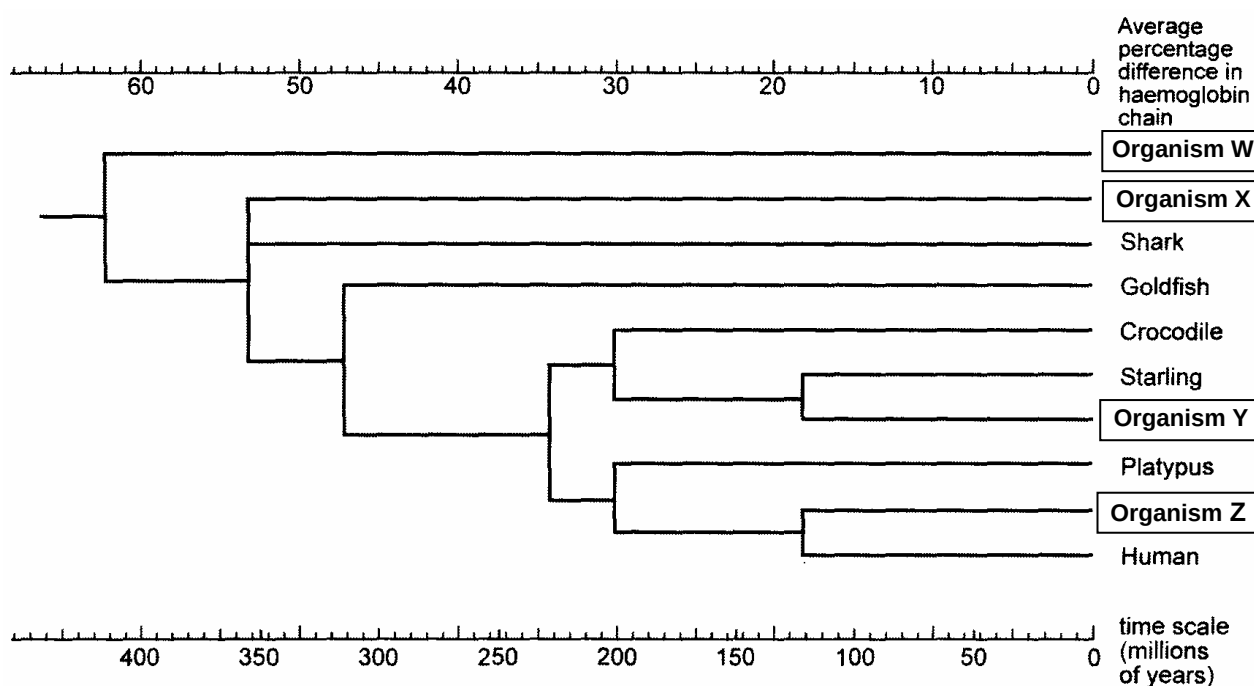


Fig. 7.2

- (a)(i) Using information from Table 7.1, identify organisms **W** to **Z**.

W
X
Y
Z

[2]

- (ii) Explain how differences in amino acid sequences in the alpha haemoglobin chain allows the establishment of a phylogenetic tree.

.....

.....

.....

.....

..... [3]

- (b) Explain the difference between classification and phylogeny.

.....

.....

..... [2]

- (c)(i) State **two** advantages of using molecular evidence compared to homology in establishing phylogeny.

.....

.....

..... [2]

- (ii) Suggest why homology still features prominently in evolutionary studies despite the advantages molecular evidence can confer.

.....

.....

..... [2]

- (d) Explain the role of neutral mutations in evolutionary studies.

.....

.....

..... [2]

[Total: 13]

Section B

Answer any **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section **(a)**, **(b)** etc., as indicated in the question.

- 8** **(a)** Describe the fluid mosaic model of membrane structure with reference to the functions of its components. [10]
- (b)** Using appropriate examples, explain the importance of membranes to eukaryotic cells. [4]
- (c)** Describe how enzymes like amylase are synthesised and subsequently secreted from a cell. (Details of transcription and translation are not required) [6]

[Total: 20]

- 9** **(a)** Compare the structure and organization of prokaryotic and eukaryotic genomes. [8]
- (b)** Describe how eukaryotes regulate gene expression at chromosomal, post transcriptional and post translational level. [12]

[Total: 20]